

13. Analytical solutions to boundary layer equations

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2}$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$$u(x, y=0) = v(x, 0) = 0$$

$$u(x, y=\infty) = u_\infty$$

$$T(x, 0) = T_s$$

$$T(x, \infty) = T_\infty$$

[Scaling analysis]

Let's assume $Pr \ll 1$

$$\left| \begin{array}{lll} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 & \Rightarrow & \frac{u_\infty}{L} \sim \frac{v_0}{\delta} \Rightarrow v_0 \sim \frac{L}{\delta} u_\infty \\ u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} & \Rightarrow & \frac{u_\infty^2}{L} \sim \nu \frac{u_\infty}{\delta^2} \Rightarrow \delta \sim \sqrt{\frac{\nu L}{u_\infty}} \\ u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} & \Rightarrow & u_\infty \frac{\Delta T}{L} \sim \alpha \frac{\Delta T}{\delta_t^2} \Rightarrow \delta_t \sim \sqrt{\frac{\alpha L}{u_\infty}} \end{array} \right.$$

- Boundary layer growth

$$\frac{\delta}{L} \sim \sqrt{\frac{\nu}{u_\infty L}} = \sqrt{\frac{1}{Re_L}} \quad \frac{\delta}{x} \sim \sqrt{\frac{1}{Re_x}} = Re_x^{-\frac{1}{2}}$$

$$\frac{\delta_t}{L} \sim \sqrt{\frac{\alpha}{u_\infty L}} = \sqrt{\frac{1}{Re_L Pr}} \quad \frac{\delta_t}{x} \sim Re_x^{-\frac{1}{2}} Pr^{-\frac{1}{2}}$$

High speed, small α , small $\nu \rightarrow$ Thin BL

Note momentum/thermal penetration $\delta \sim \sqrt{\nu t} \sim \sqrt{\nu \frac{L}{u_\infty}}$ and $\delta_t \sim \sqrt{\alpha t} \sim \sqrt{\alpha \frac{L}{u_\infty}}$ (This

is because velocity change is due to momentum/heat diffusion: BL thickness $\sim$ length over which velocity changes significantly $\sim$ momentum diffusion length)

- Shear stress and heat transfer coefficient

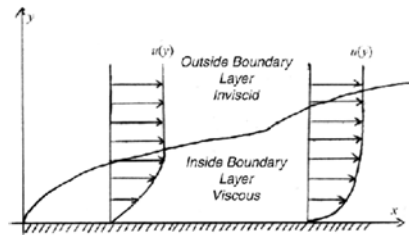
$$\tau_x = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \sim \mu \frac{u_\infty}{\delta} \sim \mu^{\frac{1}{2}} u_\infty^{\frac{3}{2}} \rho^{\frac{1}{2}} x^{-\frac{1}{2}} \quad C_{f,x} \sim Re_x^{-\frac{1}{2}}$$

$$h \sim \frac{k}{\delta_t} \sim k \sqrt{\frac{u_\infty}{\alpha x}} \sim (k \rho c_p u_\infty)^{\frac{1}{2}} x^{-\frac{1}{2}} \quad Nu_x \sim Re_x^{\frac{1}{2}} Pr^{\frac{1}{2}}$$

- ✓ Thin BL $\rightarrow$ Large velocity gradient $\rightarrow$ Large shear stress
- ✓ τ_x and $h \sim x^{-1/2}$ (BL thickness $\sim x^{1/2}$)
- ✓ Shear stress proportional to $u_\infty^{\frac{3}{2}}$ (Why? $\tau_x \sim \mu \frac{u_\infty}{\delta}$)

- **Shear stress and heat transfer coefficient**

Sketch on the following the variations of heat transfer coefficient, shear stress, boundary layers, pressure, etc. as a function of x



[Similarity solution (Blasius solution)]

The BL equations can be solved by the similarity solution method (see 7.2.1 for details)

<Main idea>

PDE $u(x,y), v(x,y), T(x,y) \rightarrow$ ODE(η)

The similarity variable η is obtained by physical intuition, dimensional analysis, etc.

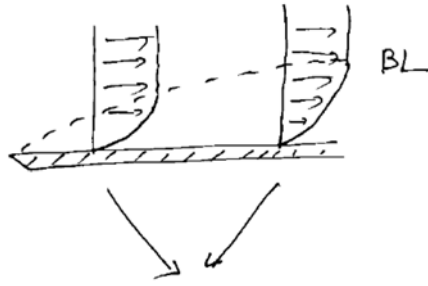
New independent variable
$$\eta \equiv y \sqrt{\frac{u_\infty}{\nu x}} = \frac{y}{\sqrt{\nu x / u_\infty}}$$

Instead of working with u, v, T , dimensionless stream function f and dimensionless

temperature T^* are used: $f \equiv \frac{\psi}{\sqrt{u_\infty \nu x}}$ and $T^* \equiv \frac{T - T_s}{T_\infty - T_s}$

BL equations then become

$$\begin{cases} 2 \frac{d^3 f}{d\eta^3} + f \frac{d^2 f}{d\eta^2} = 0 & f(0) = 0, f'(\infty) = 1 \\ \frac{d^2 T^*}{d\eta^2} + \frac{\text{Pr}}{2} f \frac{dT^*}{d\eta} = 0 & T^*(0) = 0, T^*(\infty) = 0 \end{cases}$$



These two velocity profiles are different but similar.

→ If y variable is normalized (scaled) by δ , they should be identical

→ Rewriting the G.E. using $\frac{y}{\delta} = \frac{y}{\sqrt{\frac{2x}{u_{\infty}}}} \equiv \eta$ converts the PDE into an ODE of η .

Numerical solution to the ODEs yields

$$\frac{\delta}{x} = 5.0 \text{Re}_x^{-\frac{1}{2}}$$

$$\frac{\delta}{\delta_t} \approx \text{Pr}^{\frac{1}{3}} \quad (0.6 \leq \text{Pr} \leq 60)$$

$$C_{f,x} = 0.664 \text{Re}_x^{-\frac{1}{2}}$$

$$\text{Nu}_x = 0.332 \text{Re}_x^{\frac{1}{2}} \text{Pr}^{\frac{1}{3}} \quad (0.6 \leq \text{Pr} \leq 60)$$

For air $u_{\infty} = 100 \text{ km/h}$, $\delta = 3.8 \text{ mm}$ at $x = 1 \text{ m}$ (air $\nu = 1.589 \times 10^{-5} \text{ m}^2/\text{s}$)